

Ethanol Blending in Gasoline - Panama

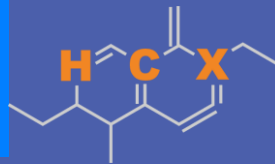
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Ethanol Blending in Latin America

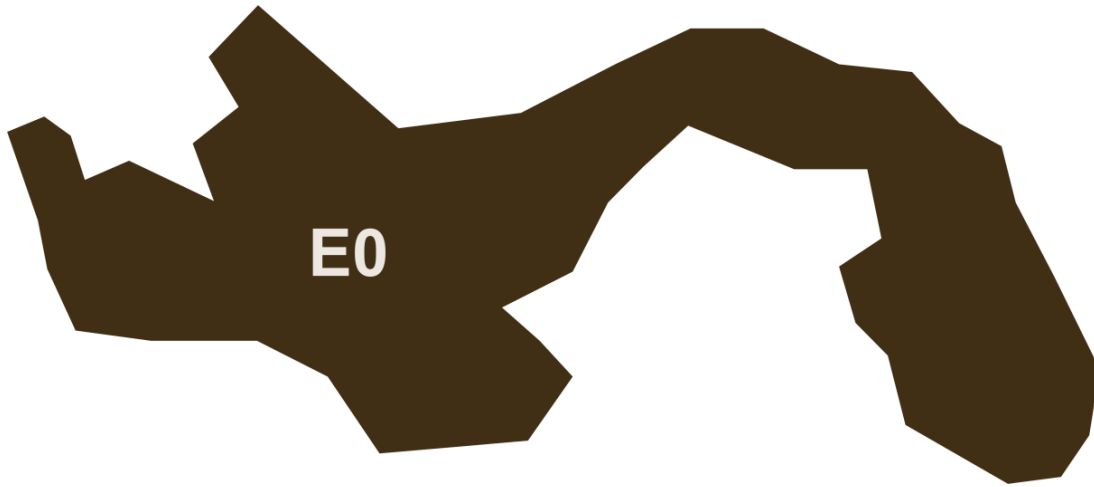
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



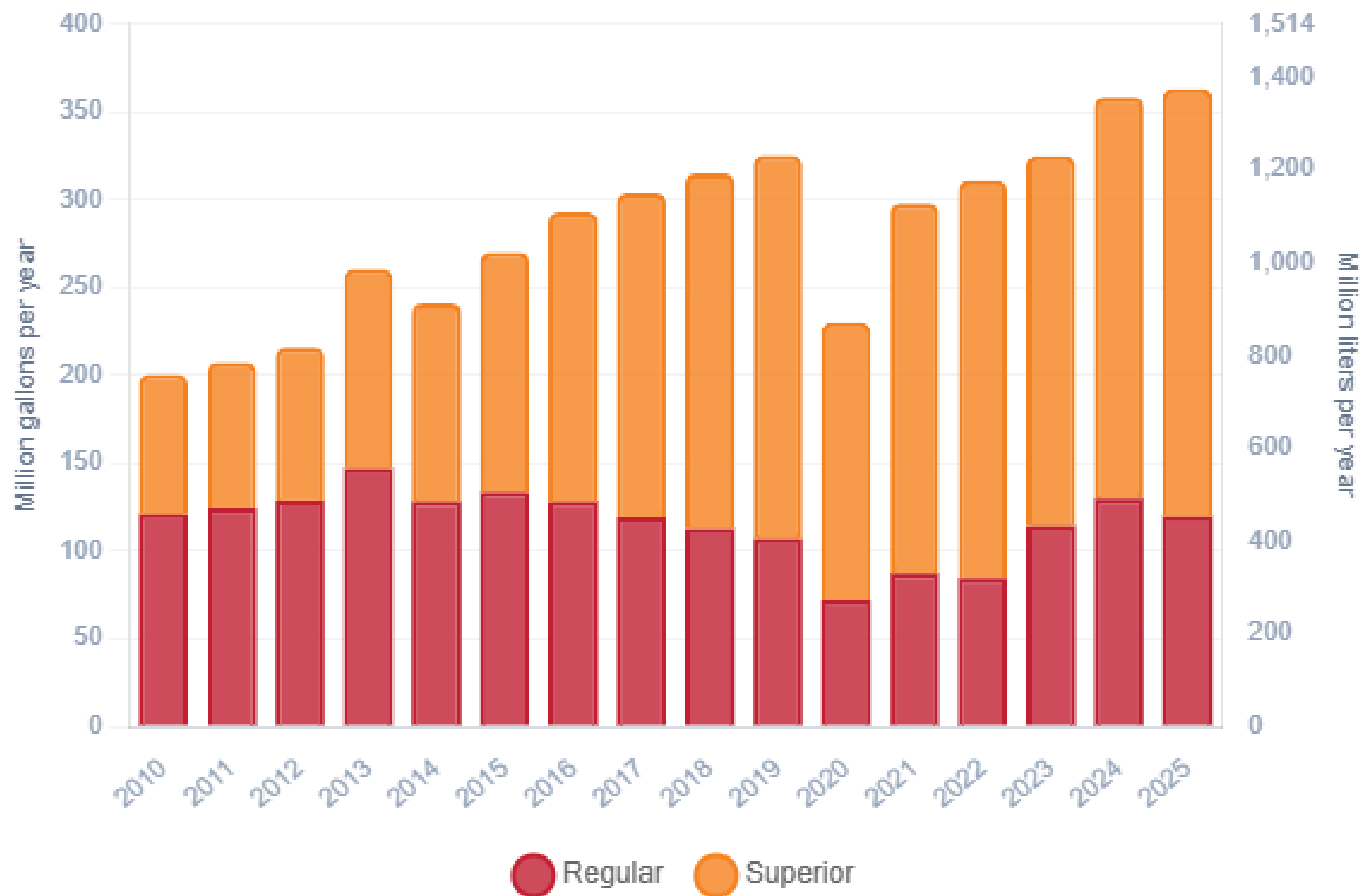
Ethanol Blending in Gasoline - Panama



En 2025 demand was 363 million gallons (1374 million liters). Regular gasoline (RON 91) represented 33% of total demand and Superior gasoline (RON 95) 67%. Gasoline is supplied only with imports, mainly from United States, the Netherlands and Germany.

In 2011, 2% v/v ethanol blends with gasoline were authorized, with a planned gradual increase until reaching 10% v/v by 2016. However, in 2013 it was limited to 5% v/v and by 2014 blending stopped because the only national producer announced its production cease due to political dispute. A new mandate implementation has been approved for E10 and is expected to be implemented between 2026 and 2027.

Gasoline Demand in Panama



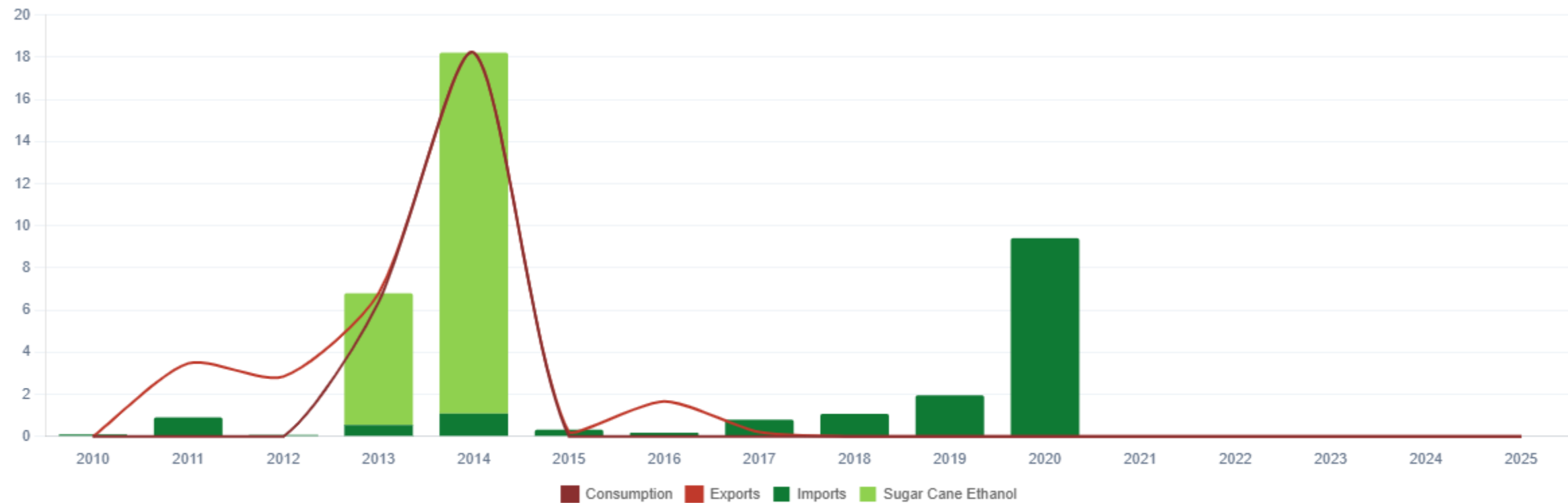
Source: Secretaría de Energía - Panama

Gasoline Imports to Supply Demand in Panama



Source: Secretaría de Energía - Panama

Ethanol Balance in Panama



Source: Secretaría de Energía - Panama

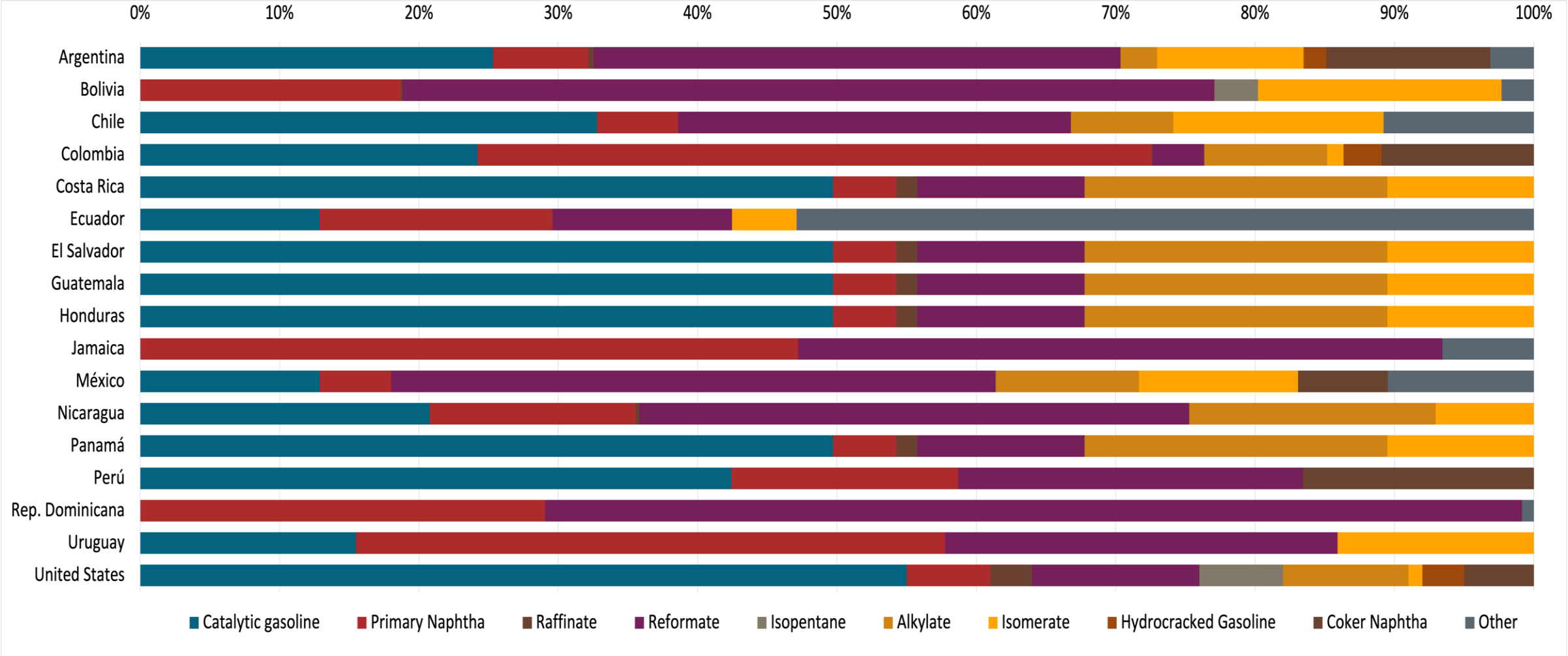
Gasoline Quality in Panama

Name	DGNTI-COPANIT 83-2013/ Resolution 425/2020		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2014/2021		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	RON 91	RON 95	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 5% v/v / <1,5% v/v	< 5% v/v / <1,5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 50% v/v	< 50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 30% v/v	< 30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	-	-	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 91	> 95	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg / < 150 mg/kg	< 500 mg/kg / < 150 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	-/ < 0,7% v/v	-/ < 0,7% v/v	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	10% v/v	10% v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa (< 76 kPa if ethanol is added)	< 69 kPa (< 76 kPa if ethanol is added)	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	0% v/v if ethanol is added	0% v/v if ethanol is added	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: RTCA

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



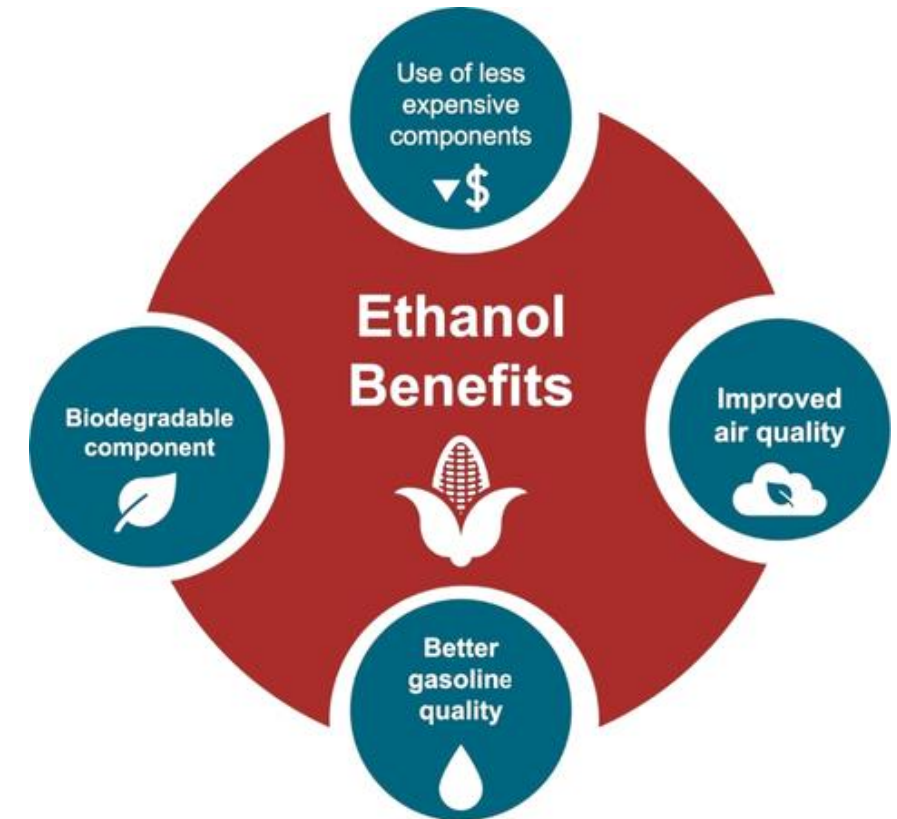
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

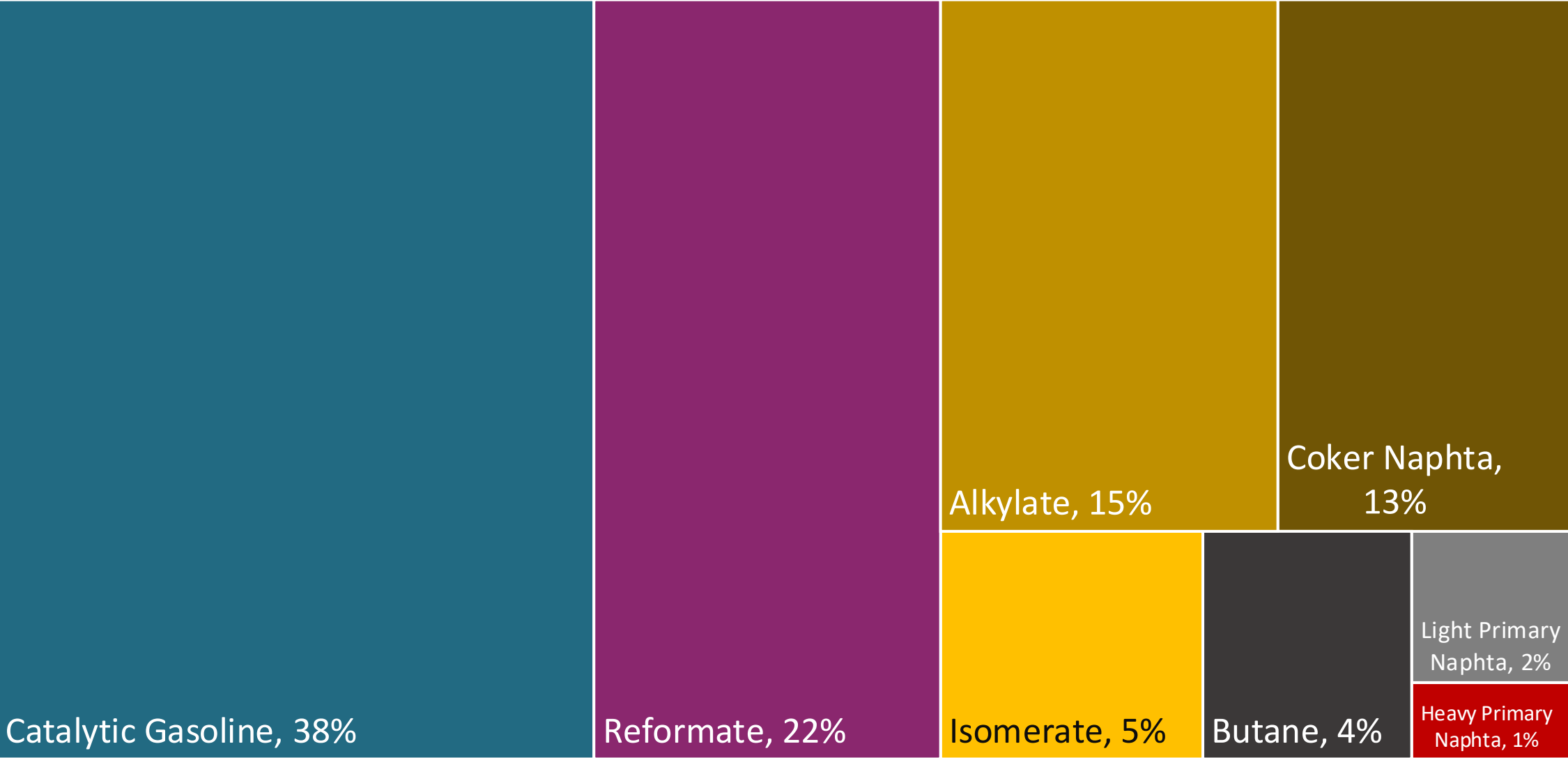
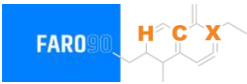
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

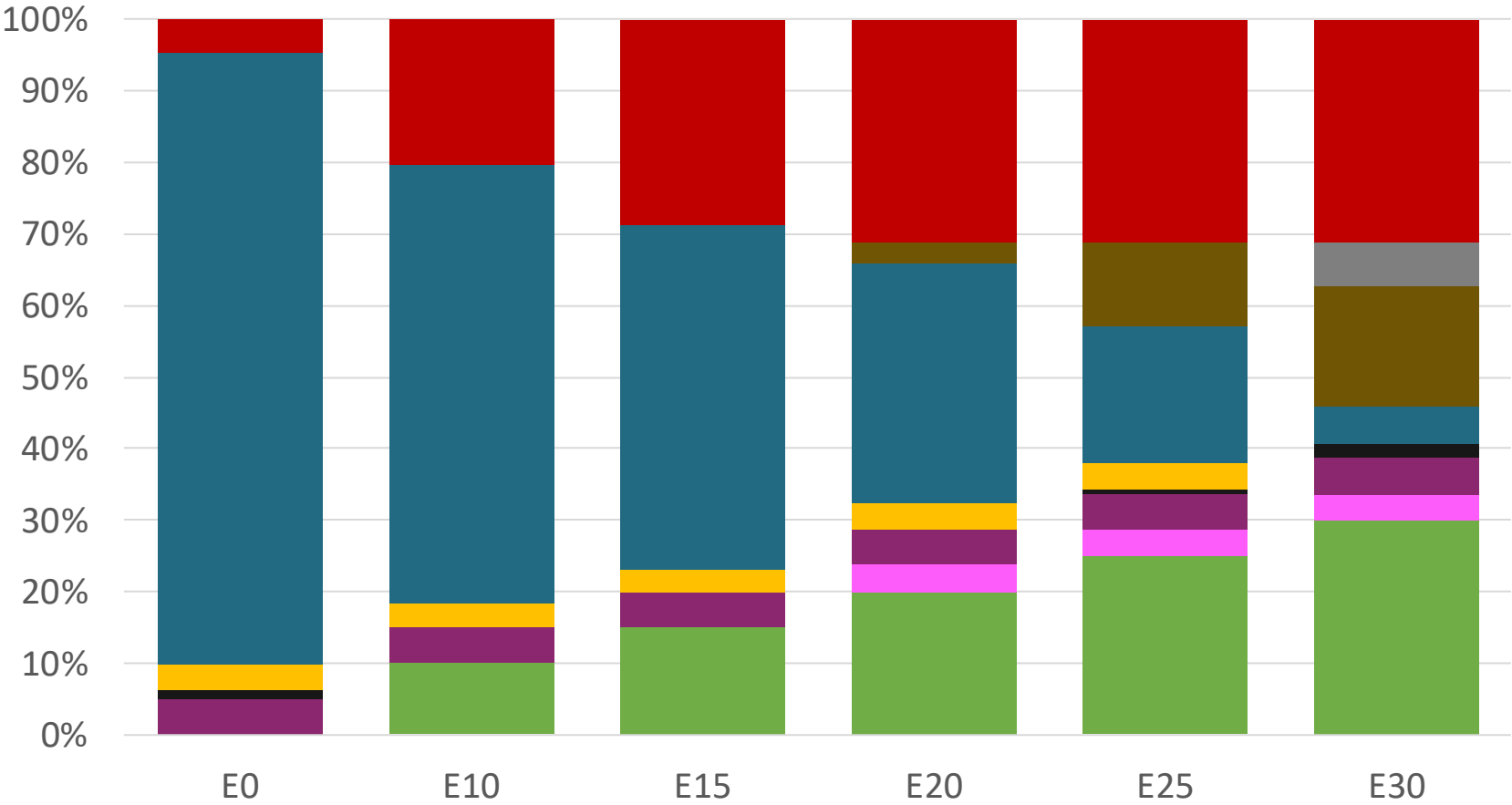
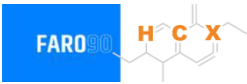


Blending Components - Panama



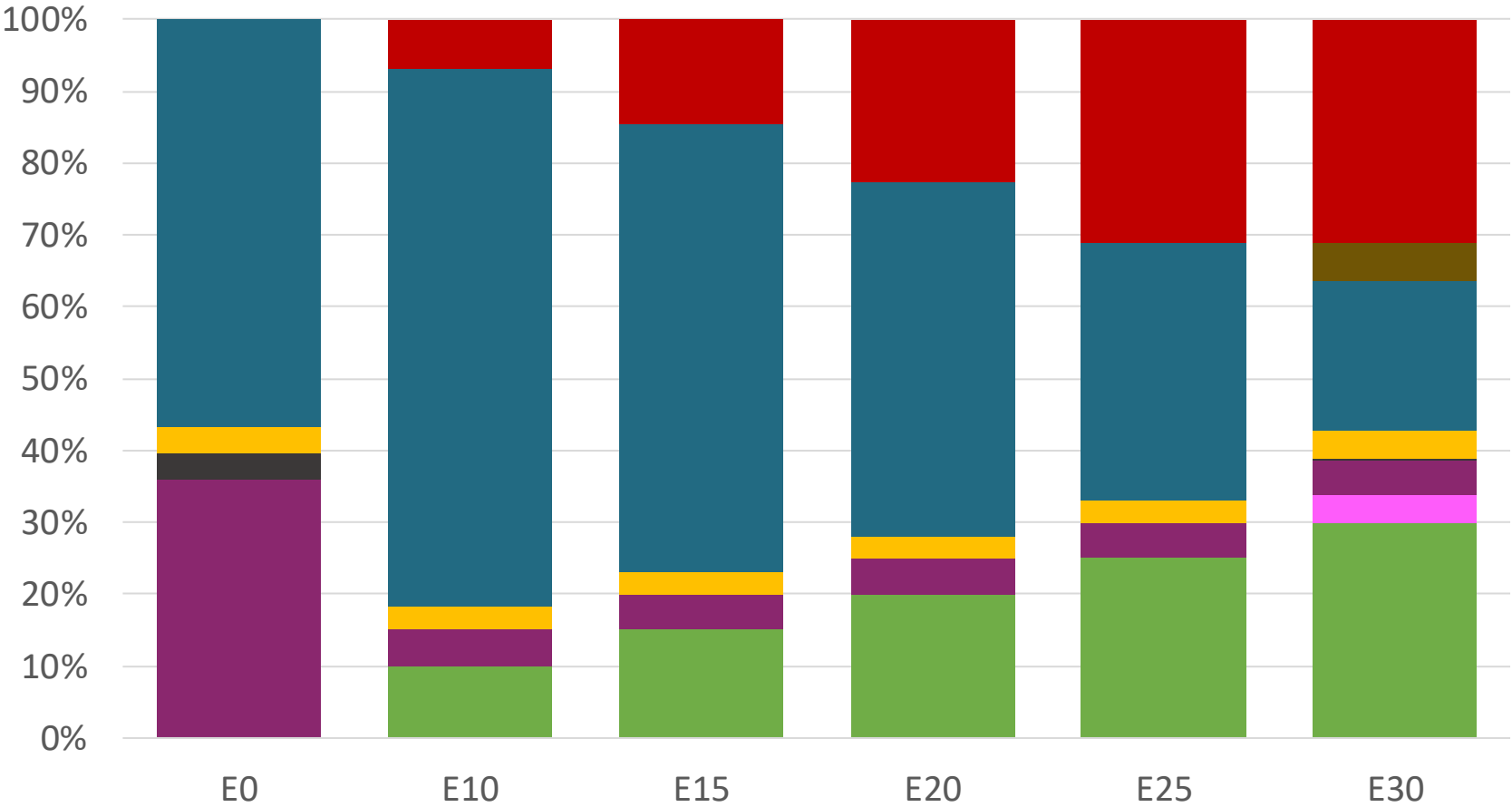
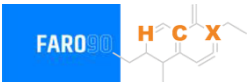
Source: Faro90

Panama – Regular – Constant Octane



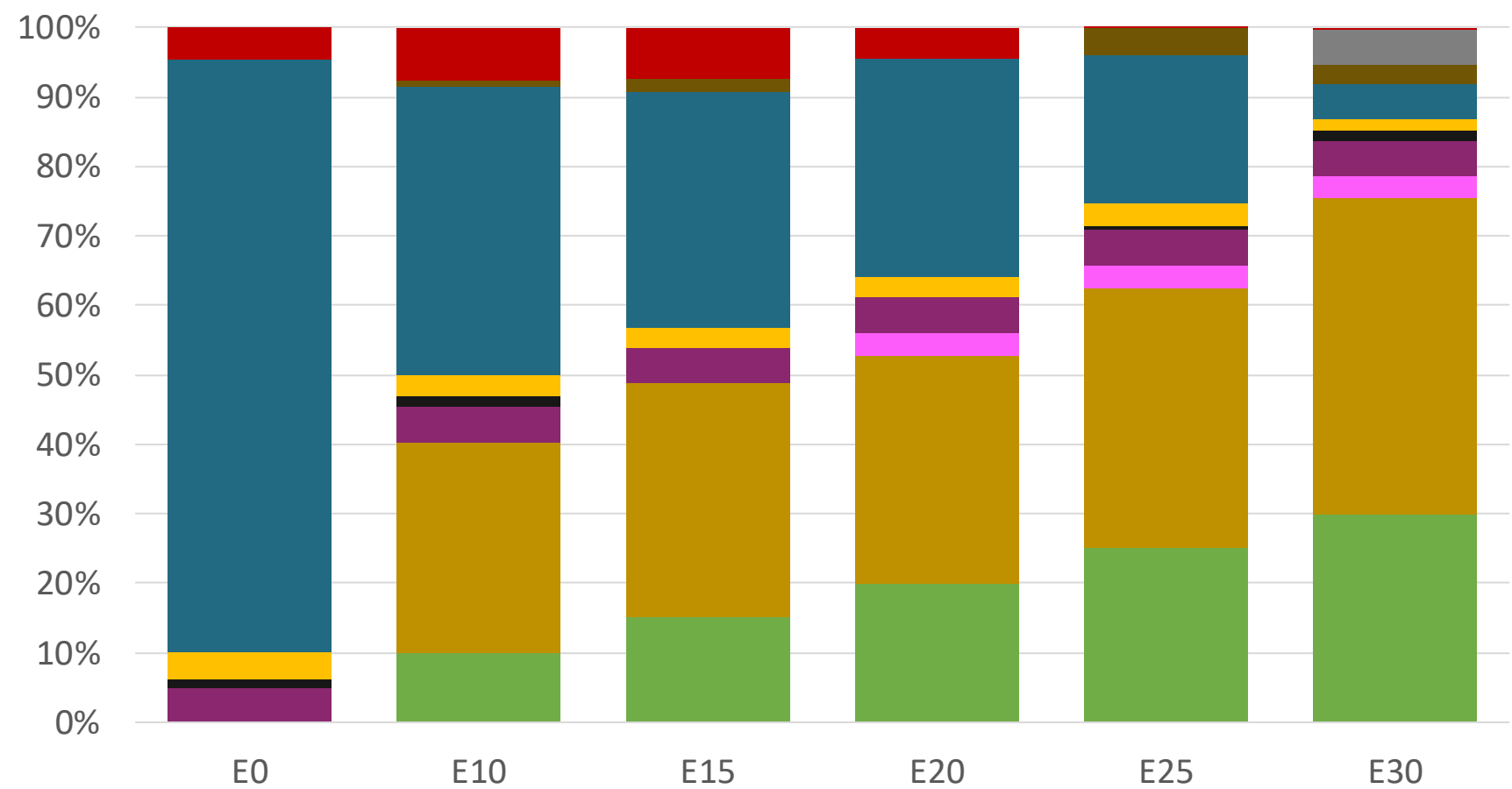
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	\$ 2.076	\$ 1.817	\$ 1.673	\$ 1.526	\$ 1.375	\$ 1.257

Panama – Premium – Constant Octane



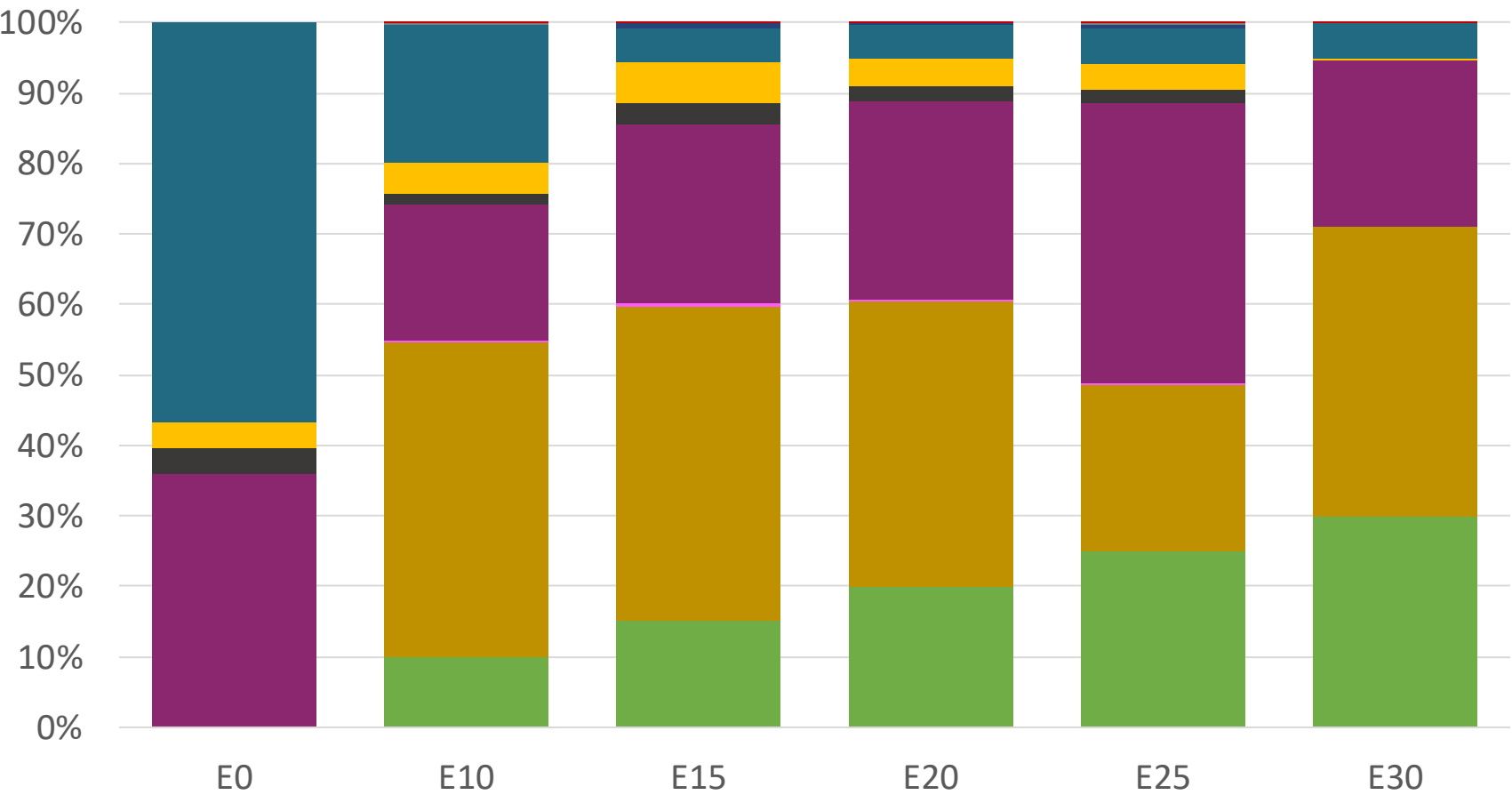
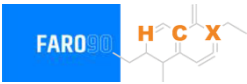
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	\$ 2.326	\$ 2.023	\$ 1.888	\$ 1.746	\$ 1.597	\$ 1.448

Panama – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	\$ 2.076	\$ 2.076	\$ 2.076	\$ 2.076	\$ 2.076	\$ 2.076

Panama – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

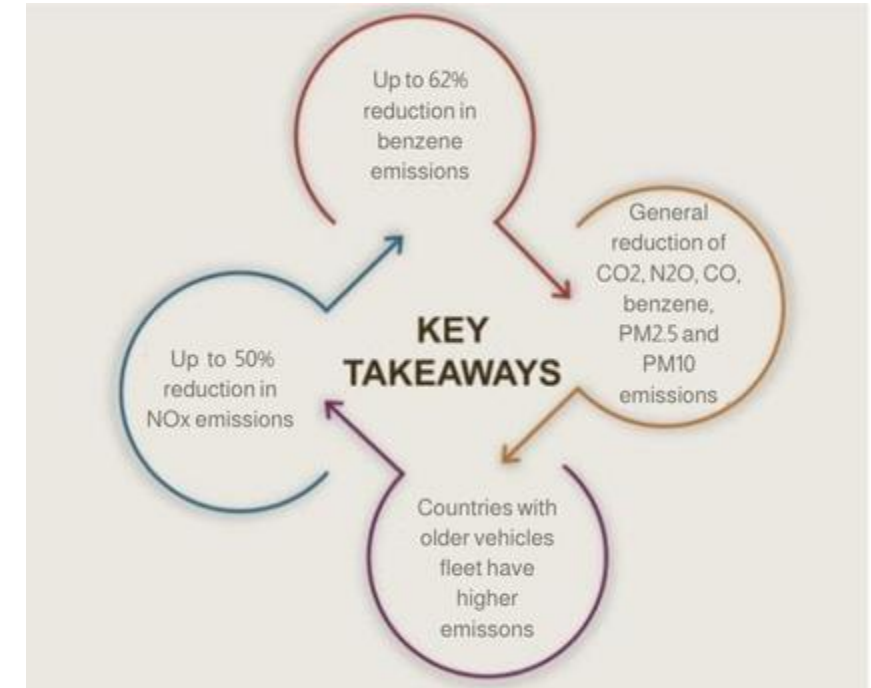
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

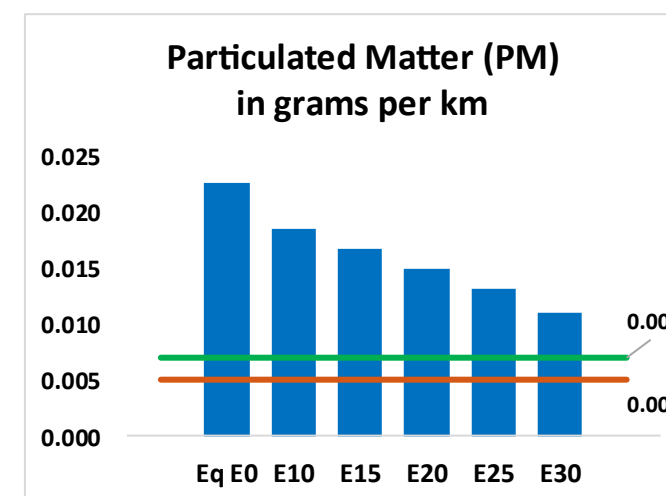
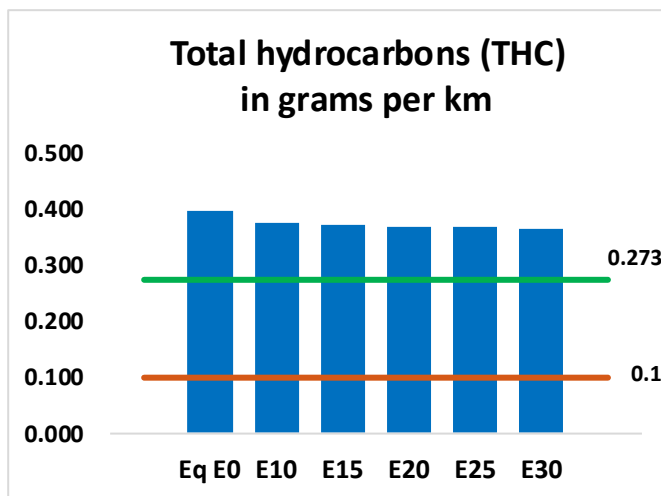
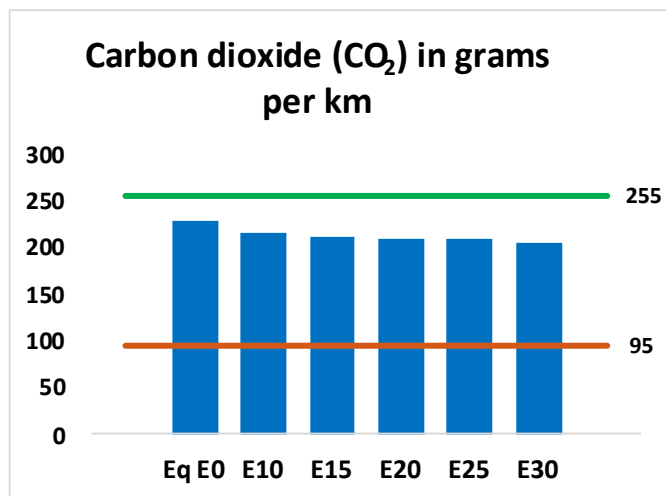
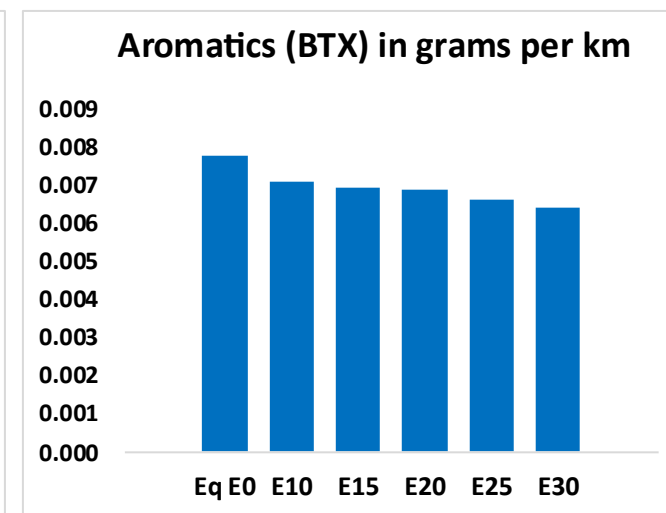
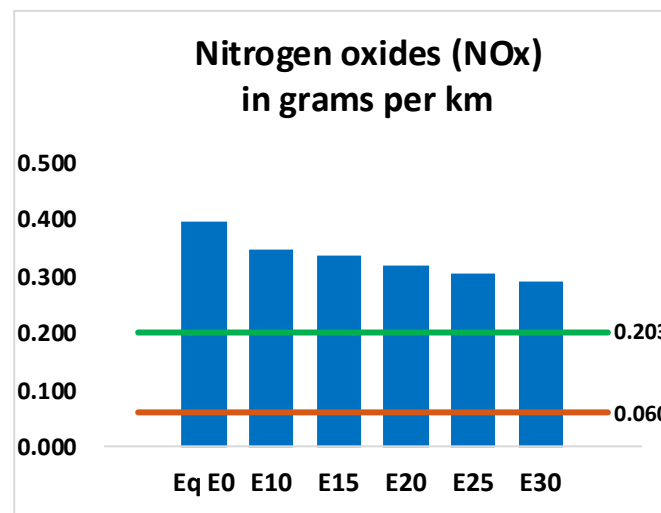
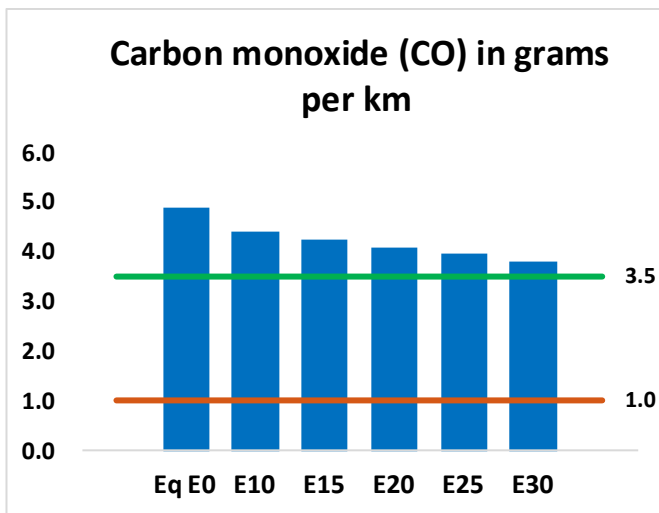
Gasoline Vehicular Fleet in Panama

Type	Engine	Exhaust	Recovery	Age	Number of vehicles				
Small automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	764				
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	5,313				
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	69,297				
		Euro III		> 8 years, > 160 tkm	36,491				
		Euro IV		> 8 years, > 160 tkm	53,388				
		Euro V / Euro 6		> 8 years, > 160 tkm	88,980				
				> 8 years, > 160 tkm					
	Hybrid			4-8 years, 80-160 tkm	60,428				
	Electric			< 4 years, < 80 tkm	49,672				
Medium automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	4,777				
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	17,469				
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	82,743				
		Euro III		> 8 years, > 160 tkm	32,962				
		Euro IV		> 8 years, > 160 tkm	53,136				
		Euro V / Euro 6		> 8 years, > 160 tkm	88,560				
				> 8 years, > 160 tkm					
	Hybrid			4-8 years, 80-160 tkm	97,068				
	Electric			< 4 years, < 80 tkm	123,287				
Small Engines	FI 4-cycles	None	PCV	> 8 years, > 160 tkm	8,716				
		Euro II		> 8 years, > 160 tkm	3,140				
		Euro III		> 8 years, > 160 tkm	19,498				
		Euro IV		> 8 years, > 160 tkm	3,828				
		Euro V		4-8 years, 80-160 tkm	6,380				
				4-8 years, 80-160 tkm	3,562				
				< 4 years, < 80 tkm	7,123				
	Electric		Any						
Heavy	Carburator	3-Ways	PCV	> 8 years, > 160 tkm	169				
	FI	3-Ways		> 8 years, > 160 tkm	23,019				
		Euro III		> 8 years, > 160 tkm	11,110				
		Euro IV		> 8 years, > 160 tkm	17,322				
		Euro V / Euro 6		> 8 years, > 160 tkm	28,870				
				> 8 years, > 160 tkm					
	Hybrid			4-8 years, 80-160 tkm	16,958				
	Electric		< 4 years, < 80 tkm	10,820					

Vehicle Fleet: **916,582**
Average Age: **11.5 years**
Motorcycles: **5%**
Source: INEC, analysis Faro 90

Vehicle Emissions in Panama

United States
Euro 6 Standard



Panama – Vehicle Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	101,418	96,500	91,582	88,113	84,881	82,474	79,107	-10%	-16%
CO2	4,716,772	4,595,483	4,480,933	4,390,843	4,346,262	4,324,891	4,245,171	-5%	-8%
VOC	8,233	8,023	7,814	7,725	7,673	7,645	7,567	-5%	-7%
NOx	8,260	7,627	7,195	6,976	6,622	6,345	6,019	-13%	-20%
SOx	56	52	48	44	40	37	33	-15%	-28%
NH3	1,545	1,530	1,514	1,521	1,539	1,551	1,561	-2%	0%
N2O	99	98	98	98	100	101	103	-1%	2%
CH4	1,793	1,793	1,793	1,820	1,859	1,893	1,926	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	99	97	96	95	95	95	95	-4%	-4%
Acetaldehyde	179	205	301	459	625	714	844	68%	249%
Formaldehyde	533	518	604	710	743	822	895	13%	39%
Benzene	161	155	146	144	143	137	132	-9%	-11%
PM2.5	262	238	214	193	173	151	128	-18%	-34%
PM10	209	190	171	154	138	120	102	-18%	-34%

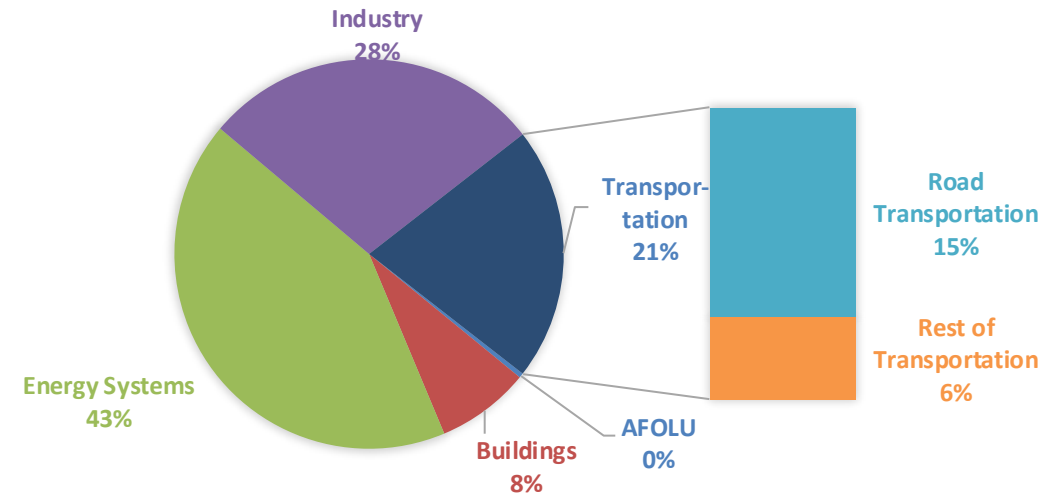
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

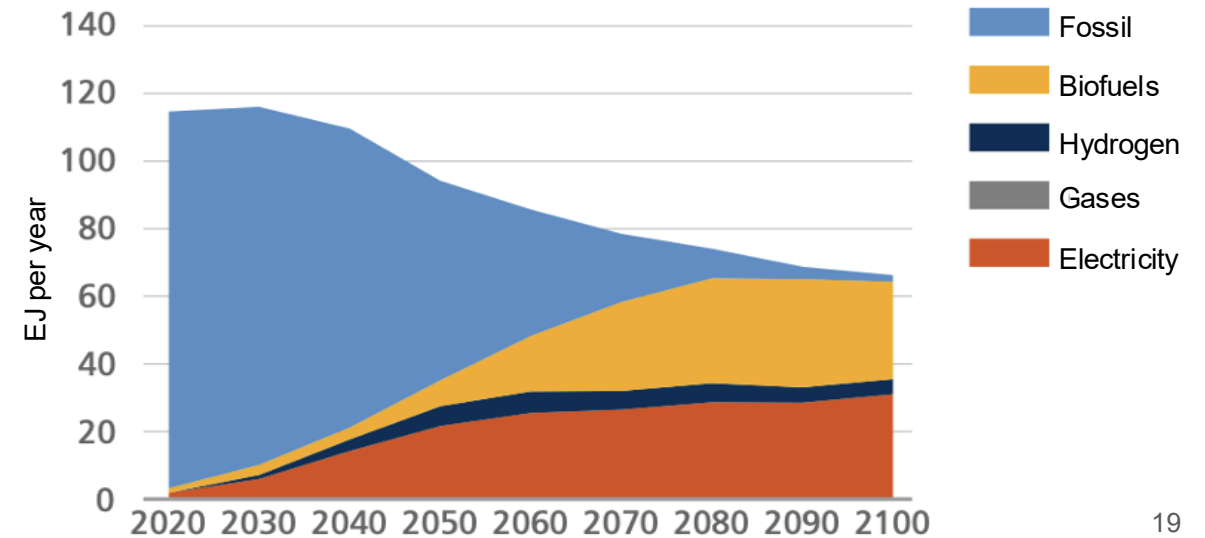
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

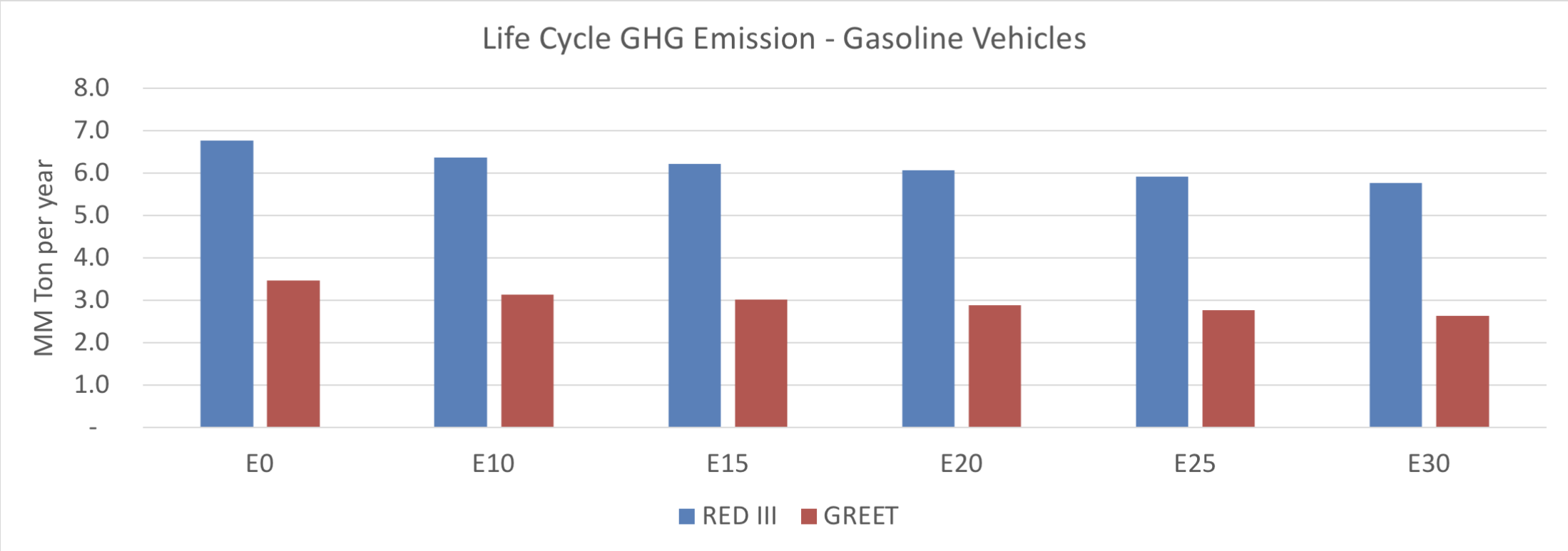
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Panama – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	23.8
2035 Target (MMT/year)	16.7
Delta	7.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.6%	13.3%	16.9%	20.1%	24.2%
RED II/III	0.0%	5.9%	8.2%	10.5%	12.5%	14.9%
% 2035 Target	E0	E10	E15	E20	E25	E30
2025	0.0%	4.7%	6.4%	8.2%	9.8%	11.8%
RED II/III	0.0%	5.6%	7.8%	9.9%	11.8%	14.1%